MATH 3042- Quiz 2- Winter 2026- Solutions

January 27, 2026 1:13 PM

MATH 3042- Quiz 2
Time: 20 minutes

Full Name:
Set:

Q1 (6 marks)

You are comparing two database clusters.

- **Cluster A:** [102, 105, 103, 104, 101]
- **Cluster B:** [80, 100, 150, 90, 95]

Which cluster has a higher **coefficient of variation**?

Provided summary of cluster A: n = 5, Mean = 103, Var = 2.5 (no need to recalculate, please use these values)

A: coeff of var = $\frac{\sqrt{2.5}}{103} = 1.5\%$

For cluster Y, manually do all the required calculations, show all work please.

Cluster B	$x - 103$	$(x-103)^2$
80	-23	529
100	-3	9
150	47	2209
90	-13	169
95	-8	64

B:

$var = \frac{2980}{4} = 745$

$S = \sqrt{745} = 27.3$

Coeff of var $= \frac{27.3}{103} = 26.5\%$

Sum 515

$\bar{x} = 103$

$$var = \frac{\sum(x-\bar{x})^2}{n-1}, \quad S = \sqrt{var}, \quad coeff\ of\ var = \frac{S}{\bar{x}}$$

Cluster B has a higher coefficient f variations

Q2 (5 marks)

Consider the following data for memory usage (MB): [200, 210, 220, 230, 240, 250, 260,1000]

1. What is the **Median**?
2. Calculate **Q1** and **Q3**.
3. Calculate the Upper fence and lower fence
4. Are there any outliers?

I: 1 2 3 4 5 6 7 8

$$IQR = Q_3 - Q_1, \quad UF = Q_3 + 1.5 \times IQR, \quad LF = Q_1 - 1.5 \times IQR$$

$$Median: \frac{230 + 240}{2} = 235$$

$$Q_1 = \frac{210 + 220}{2} = 215$$

$$Q_3 = \frac{250 + 260}{2} = 255$$

$$IQR = 255 - 215 = 40$$
$$UF = 255 + 1.5 \times 40 = 315$$
$$LF = 215 - 1.5 \times 40 = 155$$

There are no values less than 155, we do not have lower outliers
The is one value exceeding the upper fence, we have one upper outlier: 1000

Q3. (4 marks)

A server's average temperature is 50°C with a standard deviation of 4°C.

1. Calculate the **Z-score** for a temperature reading of **59°C**.
2. Using the $|z| > 2$ rule, is **59°C** considered an anomaly?
3. If a Z-score of a certain data point is -1.2, what does that tell you about that specific data point relative to the mean (interpret the given z-score)?

1. $z - score = \frac{59 - 50}{4} = +2.25$
2. Yes, the z-score is outside of ± 2 therefore $59^o C$ is an anomaly
3. It means the recorded temperature is 1.2 standard deviations below the average